

Vendor (audited): Kansai Nerolac Paints Ltd (KNP)

Customer / Location: MSIL — Paint Shop A, Kharkhoda (PS-KA)

Product: PVC Sealer (UBS & seam sealer)

Application: Robot (1 mm nozzle) + manual gun

FTIR shared with: M. Awasthi, A. Chaurasiya (MSIL)

Prepared by: [name / dept]

Concern: Recurring choking of 1 mm UBS robot nozzles with barrel-to-barrel variation. Visual check of choke material shows tiny black particles, reddish-brown flakes, barrel-gasket chips, paint chips and black chips from the polyethylene bag, retained in barrels, filters and nozzles.

KEY FINDING — KNP's comparative FTIR of the choke particles vs. *all* suspected foreign particles showed **NO IR match** — but this is likely a **false negative**: after soaking in sealer, packaging absorbs plasticizer and is coated with a sealer film (carbonate / PVC) that masks and shifts its own bands, so *virgin* packaging will never match a *soaked* flake. **NEW LEAD:** the **imported (Malaysia) RM packaging** visually matches the choke flakes and is the prime suspect. Compare **like-for-like** — FTIR the packaging *after* matched sealer-soaking, and after solvent-extracting the flake to strip the sealer coat and recover the substrate; confirm the polymer by DSC / melt-point & TGA.

A. Paint-Shop Issues — MSIL Kharkhoda (PS-KA)		
#	Issue / Observation	Status / Remark (KNP)
1	Robot nozzle choking (UBS): tiny black particles in sealer batch; barrel-to-barrel variation; solid particles, barrel-gasket chips, paint chips; PE bag with black chips.	Comparative FTIR done — no IR match to choke particles. Reports shared with MSIL.
2	Sealer flow / dripping on vehicles — manual sealer (viscosity 105 Pa.s).	Modified batch to reach MSIL 27/07/26 — Batch 4137336.
3	Sealer flow / dripping on vehicles — robot sealer/UBS (viscosity 110 Pa.s).	Modified batch to reach MSIL 27/07/26 — Batch 4145253.
4	UBS batch size 1 KL < line volume — hard to monitor batch-wise performance. Raise to 3–5 T for 3–4 days' consistency.	Open — KNP to advise on larger batch sizes after internal review.
5	Whiteness issue.	Not resolved — KNP to improve & share cut-off batch no.
6	Baking-repair cracking (heat stabilizer raised 1.1 → 1.3%).	Resolved — no crack after KNP countermeasure batch.

B. KNP Plant Audit Points		
#	Audit Point	Status / Remark
1	Hopper cleaning OK — strengthen cleaning process further.	Implemented.
2	Remove corrugated sheet from pallets.	Implemented — eliminated from process.
3	Replace wooden brush with metal brush.	Open — aluminium-handle brush by 30/07/26.
4	Remove cardboard lid of standard-weight box; use metal lid.	Implemented — metal lid in use.
5	RM packaging: move powder bags → polythene (with purchase & supplier).	Open — under exploration.
6	Increase filter-cleaning frequency 1× → 2× per week.	Implemented from 28/07/26.
7	Test self-cleaning-filter residue & compare with FG.	Open — report by KNP on 04/08/26.
8	Soak RM-bag paper pieces in sealer 24 h & test by FTIR.	Under test — report tomorrow.
9	Use polythene for sealer filling inside barrels.	Open — vendor/procurement on fast track.
10	Barrel cleaning OK — strengthen process further.	Implemented.
11	Shop-floor cleaning OK — clean sticky areas, improve.	Implemented.

- Priority open actions:**
- Test the imported-Malaysia packaging correctly:** soak a coupon in the same sealer (24–72 h, matching barrel residence) then FTIR, *and* solvent-extract the choke flake to strip the sealer coat and FTIR the recovered substrate — plus DSC / melt-point & TGA to ID the polymer independent of surface masking. Extend the RM-bag soak-FTIR test to the plastic liner.
 - Self-cleaning-filter residue vs FG report (04/08); metal brush (30/07); polythene for RM bags & FG barrel filling — fast-track.
 - Confirm larger UBS batch size (3–5 T); resolve whiteness (share cut-off batch); verify flow after modified batches 4137336 / 4145253.

Prepared / MSIL
Name: _____
Sign / Date: _____

Reviewed / MSIL Quality
Name: _____
Sign / Date: _____

Acknowledged / KNP
Name: _____
Sign / Date: _____

Prepared from joint MSIL–KNP MoM (tab: Issues), FTIR overlay and shop-floor sampling. Bracketed [] fields to be completed. The FTIR "no match" is against virgin suspected sources and is likely a soaking artifact; soak-matched / extraction-corrected FTIR plus thermal ID of the imported-Malaysia packaging is needed to confirm the source.